■■ SPECIALIZED EXPLORATION & EXOBIOLOGY FRAMEWORK

A comprehensive manual focusing on deep space orbital mechanics, biological scanning criteria, and data management tracking routes.

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1. Vessel Outfitting Rules

Exobiology requires prioritizing landing convenience, maneuverability, and shield durability over chasing pure maximum jump ranges.

- **Small Hulls (The Gold Standard):** Vessels like the Diamondback Explorer (DBX) or Dolphin excel because their tiny footprints allow them to easily squeeze onto rough, jagged mountain ledges right beside biological targets.
- **Advanced Options:** The Faulcon DeLacy Nomad or Manderlay provide ideal cockpit geometry, walkable frames, and agile thrust setups for precision low-altitude planetary flight.
- **Critical Modules:** Always install a Supercruise Overcharge (SCO) FSD to slash transit times out to distant worlds. Maintain an A-rated shield generator to absorb accidental ground impacts on high-gravity planets.
- **Artemis Suit:** Ensure your Commander is outfitted with an Artemis Suit from a Pioneer Supplies vendor storefront—its integrated Genetic Sampler tool is mandatory for exobiology logs.

2. The Deep Space Scanning Protocol

To secure the coveted **First Footfall bonus**—which scales payouts by a massive 500%—you must venture roughly 800 to 1,500 light-years completely outside the core human bubble. Set your galaxy map filters to show star types A, F, G, and K (systems holding the highest probability of thin-atmosphere worlds).

The Operational Progression Loop: Arrive in System ➔ Discovery Scan 'Honk' ➔ FSS Scanning Interface ➔ Filter Biological Signals

1. **The System Honk:** Fire your FSD Discovery Scanner ('Honk') immediately upon hyperspace exit to establish the central orbital body architecture.
2. **The Full Spectrum Scan (FSS):** Open your FSS scanner screen. Tune the spectral frequency dial along the bottom axis to locate landable planets with thin atmospheres.
3. **The Target Selection:** Check the upper-right info card for a 'Biological Signals' counter. Prioritize High Metal Content (HMC) and Ammonia worlds showing multiple biological signals.

3. Establishing Orbit & The DSS Scan

Once you identify a planet with biological signals, fly directly to it and establish a stable scanning position inside supercruise.

- **Approach Vectors:** Fly directly toward the sunny side of the planet body. This ensures full visibility for visual plant tracking once you drop into glide mode.
- **Establishing Orbit:** Throttle down your ship to minimum supercruise velocity (30 km/s) as you enter close approach. Keep your cockpit outside the yellow planetary exclusion line to prevent an emergency emergency drop into the atmosphere.
- **The Probe Launch:** Switch your cockpit HUD to Analysis Mode and activate the Detailed Surface Scanner (DSS). Aim at the planet core to launch a central probe, then shoot matching probes at the horizon edges to map the back hemisphere.

  ■ ***Video Reference:*** *Learn how to approach a planet, manage fire groups, and launch probes smoothly with the* ***Detailed Surface Scanner Setup Guide on YouTube****.*

4. Reading the Biological Heatmap Overlay

Once DSS mapping hits 100% completion, the plain planetary surface grid is replaced by an active species biological heatmap overlay.

- **Toggling the Filters:** Do not drop out of the DSS scanner view right away. Use your assigned 'Next Filter' and 'Previous Filter' keys to cycle off the default 'All' view and isolate individual, specific plant species metrics.
- **Interpreting the Teal Zones:** The map shifts structure as you cycle filters. Focus your attention entirely on the solid, light teal/blue colored fields. These zones represent matching planetary terrain properties where that specific target can grow.

- **Terrain Correlation:** The overlay shows ecological suitability, not exact plant markers. Switch your cockpit HUD back to Combat Mode occasionally on your approach to see the raw physical land geography underneath the teal overlay colors.

  ■ ***Video Reference:*** *Master navigating the overlay colors and cycling species filters using the* ***DSS Heat Map Navigation Tutorial on YouTube****.*

5. Exobiology Mechanics & Clonal Diversity

Your genetic sampler canister requires three distinct, non-clonal samples of the exact same species to compile a complete log dataset for sale.

- **The Sample Tool:** On foot, select your Genetic Sampler (Keybind 5). Hold primary fire down while aiming at the target flora to register the genetic code.
- **The Pulse Check:** Use secondary fire to emit an evolutionary ping pulse line. Look at the plant's color feedback: A Green Glow means it is far enough away to register, while a Blue Glow means it is too close to your previous scan and will be rejected.
- **Canister Constraints:** Your container can only store data logs for one plant at a time. Sampling an alternate plant family before collecting all three clonal updates will instantly erase your active sample data.

To guarantee your samples are accepted as genetically diverse, maintain these strict physical displacement distance gaps between extraction points:

Species Family	Common Terrain Type	Base Payout (CR)	First Logged (CR)
Stratum Tectonicas	Flat, open plains zones	19,010,000	**95,050,000 (Min. 100-200m away)**
Aleoida Coronamus	Sand and coarse gravel plains	14,350,000	**71,750,000 (Min. 500m+ away)**
Cactoida Vermis	Rocky formations / Canyons	11,210,000	**56,050,000 (Min. 300m+ away)**
Fungoida Bullata	Mountain ridges / Sharp slopes	7,420,000	**37,100,000 (Min. 300m+ away)**
Tussock Capillum	Open grassy fields / Dunes	6,310,000	**31,550,000 (Min. 200m+ away)**
Bacterium Cerberus	Flat muddy basins / Lakebeds	1,000,000	**5,000,000 (Min. 200m+ away)**

■ **Video Reference:** For full visual walkthroughs on extracting samples on foot, handling inventory canisters, and managing your exploration gear, study these references:

- *Review full planet mapping loops in the* ***Beginner's Guide to Exobiology on YouTube****.*
- *Master weapon shop suit selections via the* ***Elite Dangerous Starters Guide | Exobiology Tutorial on YouTube****.*
- *Learn data submission mechanics on foot inside the* ***Exobiology Step-by-Step Tutorial on YouTube****.*

6. Spansh Expressway Automation

To completely maximize your progression profit margins per hour, integrate external automated routing databases into your second flight display.

- **The Matrix Setup:** Navigate your web browser directly to the **Spansh Expressway to Exomastery Platform**. Input your live coordinates inside the Source System box and your vessel's max unladen Jump Range metric.
- **Filter Variables:** Lock down your Max Arrival Distance modifier to 2,000 ls to completely filter out slow, distant micro-cruise journeys. Slide the Minimum Value cutoff field to a strict 10,000,000 credits parameter.
- **The Yield Loop:** Spansh calculates an automated roadmap targeting premium worlds holding high-density Stratum Tectonicas, outputting a precise jump-by-jump flight log.
- **Elite Observatory Integration:** Install the third-party desktop overlay utility Elite Observatory alongside the BioInsights plugin. This background engine parses local client journal lines during your FSS sweeps, displaying an in-cockpit prediction window listing exactly what plants are growing on the planet surface before you ever initialize an orbital descent drop.

7. Terrain Navigation & Field Spotting

Once your vessel drops through planetary orbital glide lock coordinates, execute these specific flight patterns to locate elusive plant colonies visually:

- **The Overhead Cockpit Cruise:** Avoid landing blind. Flip your exploration vessel completely upside down at an alt vector of 30 to 50 meters, holding a slow speed step below 50 m/s. Look directly out through your upper cockpit canopy glass to spot clusters of flora growing across the ground below.
- **The Camera Drone Eye:** Deploy your ground car Scarab SRV. Toggle into the external camera suite, lock the camera orientation to a stationary point floating 15 meters directly above your vehicle frame, and use it as an aerial lookout eye to spot plants hidden behind sand dunes or hills.
- **Terrain Profiles:** Remember that *Bacteria* appears as wide, flat, oily reflections against dark clay flats or dried basins; *Fungoida* grows strictly inside jagged mountain cliffs; and premium *Stratum* blooms exclusively across flat, smooth plain sectors.

8. Powerplay 2.0 Loyalty Synergy

You can leverage galactic political allegiances to secure massive payout multipliers on your raw exploration data before ever selling them off to a local starport terminal.

- **The Pledging Strategy:** Open your system power interface and pledge your loyalty to the independent leader **Pranav Antal**.
- **The Rank Incentive:** Complete strategic faction operations (such as data hacking station advertisement boards with Recon Limpets for 21 Merits per node) to steadily build up your faction Loyalty Rank tier.
- **The Payout Stacking:** Upon securing your high loyalty rank tier, Pranav Antal awards a permanent, system-wide **+30% profit credit bonus on all organic exobiology data logs turned in** across his faction

control hubs. This passive multiplier combines with your baseline first discovery logs, turning a 95 Million credit Stratum find into a massive **123.5 Million Credit payout** per world.

■■ CRITICAL FLIGHT SAFETY WARNING - VOLATILE DATA CORRUPT RISK

Exploration logs and organic genetic structures are highly volatile assets. Unlike market trading or combat bounty milestones, your gathered scanning data **is not permanently saved to your commander file until you manually sell them to a Vista Genomics representative**. If your vessel experience hull collapse from a high-gravity terrain collision, a planetary landing mistake, or an exclusion zone drop near a neutron star, **every single unit of your unsold data is completely wiped out and destroyed**. Fly defensively, maintain full power parameters to your shield systems, and head back toward secure bubble outposts or friendly fleet carriers to cash in your records regularly.